Physical Models of BikeToolsAnalyzer

Cycling power balance and aerodynamic-drag (C_dA) estimation,
and a two-node model of core and skin temperature

Technical documentation

`src/physics/aerodyn.py` and `src/physics/thermal.py`

29 October 2026

Abstract

This report documents, derives and critically assesses the two physical models used by BikeToolsAnalyzer to interpret a recorded ride: (i) the *road-cycling power balance*, which closes the balance between measured crank power and the sum of aerodynamic, rolling, gravitational and inertial power, and from which the effective drag area C_dA is estimated by least squares; and (ii) a lumped *two-node thermoregulation model* (core and skin) that predicts core temperature from power, ambient temperature, humidity, air speed and clothing. For each model we state the governing equations, relate every term to the code, give the provenance of each constant, and quantify the behaviour of the implementation with controlled numerical experiments (Monte-Carlo recovery of a known C_dA, first-order error propagation, step-size convergence, energy-conservation checks and parameter sensitivity). The audit identified and corrected several defects, which are listed in a findings register together with the limitations that remain. The main conclusions are: the least-squares C_dA estimator is unbiased where the per-sample-ratio estimator is biased by up to 10 %, but its accuracy is limited by *systematic* errors (grade, wind, rolling resistance) of the order 0.01–0.04 m^2 rather than by random noise, and descents steeper than 5 % are excluded because unmeasured braking there would otherwise be booked as drag (+15% in a test ride). The thermal model is physically consistent and energy-conserving but *uncalibrated*, with a strong dependence on the assumed gross efficiency (its exercise set point is driven by the rider's FTP), so it should be used for relative comparisons rather than as an absolute predictor of core temperature.

Contents

1 Scope and conventions **3**

I Power balance and drag-area estimation **3**

2 Equation of motion **3**
2.1 Air speed in the presence of wind 4
2.2 Drivetrain, rolling resistance, mass and inertia 4

3 Air density **5**

4 Estimating C_dA from a ride **6**
4.1 Residual aerodynamic power 6
4.2 Pre-processing: smoothing and differentiation 6
4.3 Why least squares: the estimator 6
4.4 Exclusion of steep descents 7
4.5 Identifiability 7

4.6 Parameters . . . 7

5 Numerical verification **8**
5.1 Recovery of a known C_dA (Monte Carlo) . . . 8
5.2 Systematic errors . . . 9
5.3 Braking on steep descents . . . 10

6 Limitations of the C_dA model (not corrected in the code) **10**

II Two-node model of core and skin temperature **10**

7 Model structure **10**
7.1 Body surface area . . . 11

8 Heat production **11**

9 Heat exchange with the environment **12**
9.1 Dry exchange through the clothing . . . 12
9.2 Evaporative heat loss . . . 13
9.3 Respiratory heat loss . . . 14
9.4 Core–skin transfer and vasodilation . . . 14
9.5 Exercise set point . . . 14

10 Numerical integration **15**

11 Verification and behaviour **16**
11.1 Energy conservation . . . 16
11.2 Resting state . . . 16
11.3 Exercise scenarios . . . 16
11.4 Effect of the exercise set point . . . 17
11.5 Parameter sensitivity . . . 18

12 Limitations of the thermal model **19**

III Audit summary **19**

13 Findings register **19**

14 Recommendations **20**

A Parameters of the thermal model **21**

B Reproducibility **21**

1 Scope and conventions

The software reads a ride as a time series sampled at (nominally) 1 Hz: ground speed v, power P, grade, altitude, heading (bearing), and — from a weather service (Zippenfenig, 2023) — temperature, relative humidity, wind speed at 10 m and the direction *from which* the wind blows. Symbols are summarised in Table 1. Throughout, SI units are used; temperatures are in °C, and "grade" is the slope $s = \tan\theta$ (a grade of 5 % is $s = 0.05$).

Symbol	Meaning	Unit	Code name
P	power measured at the crank/pedals	W	`power`
λ	drivetrain loss fraction	—	`DRIVETRAIN_LOSS`
m	total mass (rider + bike)	kg	`rider_mass`, `BIKE_MASS_KG`
v	ground speed (smoothed)	m/s	`speed`
v_a	air speed relative to rider, *along* the road	m/s	`_air_speed`
C_dA	drag area (drag coefficient × frontal area)	m^2	`cda`
C_{rr}	rolling-resistance coefficient	—	`CRR`
ρ	air density	kg/m^3	`get_air_density`
s, θ	grade ($\tan\theta$), road angle	—	`grade_smooth`
β, δ, w	heading, wind-*from* direction, wind speed at rider height	°, °, m/s	`bearing, wind_deg, wind_speed`
T_c, T_s, T_a	core, skin and air temperature	°C	`core_temp, skin_temp, temp_c`
A	body surface area (Du Bois)	m^2	`du_bois_BSA`

Table 1. Principal symbols.

All numerical results of the experiments in this report (Monte-Carlo runs, sensitivities, scenario tables, convergence tests) were generated with the repository code itself; the scripts are archived with the report (Appendix B). Constants and literature values quoted from the references are identified as such.

Part I
Power balance and drag-area estimation

2 Equation of motion

Consider a rider and bicycle of total mass m moving at ground speed v along a road inclined at angle θ. Newton's second law along the road gives

$$F_{\text{drive}} = F_{\text{aero}} + F_{\text{roll}} + F_{\text{grav}} + m_e \frac{\mathrm{d}v}{\mathrm{d}t}, \tag{1}$$

with $F_{\text{roll}} = C_{rr}\, mg\cos\theta$, $F_{\text{grav}} = mg\sin\theta$, and m_e the effective mass including the rotational inertia of the wheels, $m_e = m + \sum I_w / r_w^2$. Multiplying by the ground speed v — the speed at which the driving force does work — gives the power balance (Martin et al., 1998)

$$(1-\lambda)\,P \;=\; \underbrace{\tfrac{1}{2}\rho\, C_dA\, v_a|v_a|\, v}_{P_{\text{aero}}} \;+\; \underbrace{C_{rr}\, mg\cos\theta\, v}_{P_{\text{roll}}} \;+\; \underbrace{mg\sin\theta\, v}_{P_{\text{grav}}} \;+\; \underbrace{m_e\, v\,\frac{\mathrm{d}v}{\mathrm{d}t}}_{P_{\text{accel}}}. \tag{2}$$

Three points deserve emphasis because they are common sources of error.

1. *The aerodynamic power is the drag force times the ground speed*, $P_{\text{aero}} = F_{\text{aero}}\, v$, whereas the drag force depends on the *air* speed, $F_{\text{aero}} = \tfrac{1}{2}\rho C_d A\, v_a|v_a|$. Writing $\tfrac{1}{2}\rho C_d A v_a^3$ is only correct in still air.
2. With grade $s = \tan\theta$: $\sin\theta = s\cos\theta = s/\sqrt{1+s^2}$ and $\cos\theta = 1/\sqrt{1+s^2}$. The code uses these exact expressions rather than the small-angle forms.
3. The measured power P is divided by the drivetrain: only the fraction $(1-\lambda)$ reaches the rear wheel.

2.1 Air speed in the presence of wind

Let the road direction be the unit vector $\hat{e} = (\sin\beta, \cos\beta)$ (east, north), where β is the compass bearing of travel, and let the wind blow *from* compass direction δ, i.e. the wind velocity is $\boldsymbol{w} = -w(\sin\delta, \cos\delta)$. The air velocity relative to the rider is $\boldsymbol{u} = \boldsymbol{w} - v\hat{e}$. The component of $\boldsymbol{u}$ opposing the motion (the head-wind component added to the ground speed) and the component across the road are

$$v_a = -\boldsymbol{u}\cdot\hat{e} = v + w\cos(\delta-\beta), \qquad v_\perp = w\,\sin(\delta-\beta). \tag{3}$$

A wind from straight ahead ($\delta = \beta$) therefore adds to v_a, a tail wind ($\delta = \beta + 180°$) subtracts from it. Equation (3) is exactly what `_wind_components` implements. The *magnitude* of the apparent wind, $\sqrt{v_a^2 + v_\perp^2}$, is the quantity that cools the skin; it is exposed as `apparent_air_speed` and is shared with the thermal model (Part II), so that both models use one consistent description of the wind.

Wind at rider height. Weather services provide wind at 10 m (Zippenfenig, 2023), whereas a cyclist's body presents its area at roughly 1–1.5 m. The code multiplies the 10 m wind by $k_w = 0.7$ (`WIND_HEIGHT_FACTOR`). This is consistent with standard boundary-layer descriptions (Counihan, 1975; Panofsky and Dutton, 1984): for a power-law profile $u(z)/u(10) = (z/10)^\alpha$ with $z = 1.5$ m, $\alpha = 1/7$, 0.20, 0.25 give 0.76, 0.68, 0.62; for a neutral logarithmic profile with roughness length $z_0 = 0.01$, 0.03, 0.10 m (short grass to low crops) one obtains 0.73, 0.67, 0.59. The constant 0.7 is therefore a reasonable open-country value; it is too high in built-up or wooded terrain, where hedges and buildings shelter the rider further. Because the wind enters P_{aero} with the square of v_a, this single factor is one of the larger systematic uncertainties of the model (Section 5.2).

Sign of the drag term. Equation (2) contains $v_a|v_a|$: if the tail wind exceeds the ground speed ($v_a < 0$) the drag *assists*. The implementation uses v_a^2 (through `abs`), which is exact for $v_a \geq 0$ and mis-signs the term only for tail winds stronger than the rider's speed. Since C_dA is only estimated above 5 m/s this requires a tail wind above 5 m/s at rider height (about 7 m/s at 10 m); such samples are rare but not impossible, and are listed as a limitation (Table 11).

Yaw. The drag area of a cyclist depends on the yaw angle $\psi = \arctan(v_\perp/v_a)$ between the relative wind and the direction of travel (Crouch et al., 2017). At 10 m/s with a 3 m/s cross wind $\psi \approx 17°$. The model ignores this dependence, so the fitted value is an *effective* C_dA at the mean yaw of the ride; it is not directly comparable with wind-tunnel values measured at 0°.

2.2 Drivetrain, rolling resistance, mass and inertia

- *Drivetrain loss* $\lambda = 0.03$. Chain-drive losses of the order of 2–5 % are reported (Martin et al., 1998; Spicer et al., 2001); λ depends on load, gearing and chain condition. For a power meter that measures at the rear hub, λ should be set to zero.

- *Rolling resistance* $C_{rr} = 0.005$, a typical road-tyre value on smooth asphalt (field values from about 0.003 to 0.008 are commonly reported, depending on tyre, pressure and surface). The model treats it as constant and independent of speed.
- *Mass.* The rider's mass is the profile weight; the bicycle is fixed at 8 kg. Clothing, shoes, helmet and bottles are not included. Section 5.2 shows that a 2 kg error in mass changes the fitted C_dA by less than 0.001 m^2, so this simplification is harmless for C_dA; it matters more for the climbing model, where gravity dominates.
- *Wheel inertia.* m_e is replaced by m; the neglected $\sum I_w/r_w^2$ is of order 0.1–0.2 kg for a pair of road wheels (Martin et al., 1998), i.e. about 0.2 % of the total mass, and affects only the (zero-mean) acceleration term.

3 Air density

Aerodynamic power is proportional to ρ, so ρ must be tracked with altitude, temperature and humidity. The implementation (`get_air_density`) follows three steps.

1. Pressure. From the International Standard Atmosphere (International Organization for Standardization, 1975), for altitude h (metres),

$$p(h) = p_0 \left(1 - \frac{L\,h}{T_0}\right)^{g/(R_d L)}, \qquad p_0 = 101\,325\,\text{Pa},\ T_0 = 288.15\,\text{K},\ L = 0.0065\,\text{K/m}, \tag{4}$$

with exponent $g/(R_dL) = 9.80665/(287.058 \times 0.0065) = 5.2558$. The actual station pressure differs from the ISA value by weather systems (typically ±1–3 %); this is an irreducible uncertainty unless pressure data are supplied (Table 5: +3% in ρ shifts C_dA by -0.008 m^2).

2. Vapour pressure. The saturation vapour pressure is given by the Magnus form with the coefficients of Alduchov and Eskridge (1996),

$$e_s(T) = 610.94\,\text{Pa}\ \exp\!\left(\frac{17.625\,T}{T + 243.04}\right), \qquad e = \varphi\, e_s(T_a), \tag{5}$$

where $\varphi \in [0, 1]$ is the relative humidity and T is in °C.

3. Moist-air mixture. Treating dry air and water vapour as ideal gases (Dalton's law),

$$\rho = \frac{p - e}{R_d\,T_K} + \frac{e}{R_v\,T_K}, \qquad R_d = 287.058,\ R_v = 461.495\,\text{J/(kg K)}. \tag{6}$$

The omitted real-gas compressibility correction of the CIPM-2007 formula (Picard et al., 2008) is of order 4×10^{-4} and negligible here. Table 2 lists representative values. Humidity lowers ρ because water vapour is lighter than the air it displaces; the effect is small in cool conditions (-0.3% at 15 °C, 50 %) but reaches about -1.3% at 30 °C, 80 %.

Altitude (m)	T_a (°C)	RH (%)	ρ (kg/m^3)
0	15	0	1.2250
0	15	50	1.2211
0	30	80	1.1496
1000	20	50	1.0628
2000	10	50	0.9752

Table 2. Air density computed by the code. The first row reproduces the ISA sea-level value.

4 Estimating C_dA from a ride

4.1 Residual aerodynamic power

Rearranging Eq. (2), the power that must have been spent against the air is the *residual*

$$\hat{P}_{\text{aero}}(t) = (1-\lambda)P(t) - P_{\text{roll}}(t) - P_{\text{grav}}(t) - P_{\text{accel}}(t), \tag{7}$$

and since $P_{\text{aero}} = C_dA\,k(t)$ with the known *drag-power kernel*

$$k(t) = \tfrac{1}{2}\,\rho(t)\,v_a(t)^2\,v(t), \tag{8}$$

each sample provides one equation $\hat{P}_{\text{aero},i} = C_dA\,k_i + \varepsilon_i$. Every error in the other terms is absorbed in ε_i and therefore ends up in the estimate; the quality of the result is set by how well rolling, gravitational and inertial power are known.

4.2 Pre-processing: smoothing and differentiation

Raw 1 Hz speed is too noisy to differentiate: with white noise of standard deviation σ_v, a central difference has variance $\sigma_v^2/2$ (a 0.15 m/s noise already gives an acceleration noise of 0.1 m/s^2, i.e. about 80 W of spurious P_{accel} at 9.5 m/s and 80 kg). The implementation therefore uses

$$\bar{v}_i = \tfrac{1}{5}\sum_{j=-2}^{2} v_{i+j}, \qquad a_i = \tfrac{1}{2}(\bar{v}_{i+1} - \bar{v}_{i-1}) \;\; (\texttt{np.gradient}). \tag{9}$$

Here the two moving-average windows overlap in three samples, so $a_i = \frac{1}{10}(v_{i+2} + v_{i+3} - v_{i-2} - v_{i-3})$ and $\operatorname{Var} a_i = \sigma_v^2/25$: a variance reduction by a factor 12.5 (standard deviation 3.5×). The cost is a low-pass response: a sinusoidal speed variation of period 20 s is attenuated to $\sin(5\pi f)/(5\sin\pi f) = 0.90$ of its amplitude (frequency $f = 0.05$ cycles per sample) and one of period 10 s to 0.65. Since the acceleration term has zero mean over a ride and its low-frequency content is retained, this trade-off is favourable; it introduced a bias of $+0.0003$ m^2 (0.1%) in the noise-free test of Section 5.1.

4.3 Why least squares: the estimator

Define $\mathcal{S} = \{i : \bar{v}_i > v_{\min} \text{ and } s_i \geq -s_d\}$ with $v_{\min} = 5$ m/s (below this, drag is a small fraction of the total and indistinguishable from noise) and $s_d = 5\,\%$ (`MAX_DESCENT_GRADE_PCT`): descents steeper than 5 % are excluded, for the reason given below. The fitted drag area minimises the squared residual power,

$$\widehat{C_dA} = \arg\min_c \sum_{i\in\mathcal{S}} \left(\hat{P}_{\text{aero},i} - c\,k_i\right)^2 = \frac{\sum_{i\in\mathcal{S}} \hat{P}_{\text{aero},i}\,k_i}{\sum_{i\in\mathcal{S}} k_i^2}, \tag{10}$$

the ordinary least-squares slope of a regression through the origin (Seber and Lee, 2003). If the residuals are zero-mean, uncorrelated, with variance σ^2, then

$$\operatorname{Var}\left(\widehat{C_dA}\right) = \frac{\sigma^2}{\sum_{i\in\mathcal{S}} k_i^2}. \tag{11}$$

The alternative, averaging the per-sample ratios $r_i = \hat{P}_{\text{aero},i}/k_i$, is inferior for two reasons. First, $r_i = C_dA + \varepsilon_i/k_i$ has variance σ^2/k_i^2: slow samples (small $k \propto v^3$) are extremely noisy, but the arithmetic mean weights them equally. Second, ratios with $r_i < 0$ or $r_i > 1$ are discarded as implausible; with large noise this truncation removes mostly the negative tail, so the surviving mean is *biased upward*. Equation (10) weights each sample by k_i^2, i.e. by its signal-to-noise ratio, and needs no truncation. The estimate is accepted only if at least 120 samples qualify and $\widehat{C_dA} \in [0.10, 0.80]$ m^2 (`CDA_VALID`); outside this window the value signals bad data (e.g. a faulty power meter), not a rider. The per-sample series r_i is still computed and plotted for diagnostic purposes (`calculate_CdA`), but it is not used for the headline value.

4.4 Exclusion of steep descents

Equation (2) contains no braking term: the energy dissipated by the brakes (and by cornering) is not measured by any sensor. On a steep descent the rider typically stops pedalling and brakes to control speed; the balance then attributes the whole gravitational power surplus that is not spent on rolling and drag to "drag" — the residual $\hat{P}_{\text{aero}}$ exceeds the true aerodynamic power by the brake power P_{brake}. Because the fit weights grow like $k_i^2 \propto v^6$ (Eq. (10)), the fast samples of descents carry a disproportionate share of the weight, so this bias is amplified exactly where it is most likely. Samples with grade below -5% (smoothed grade) are therefore dropped from the estimate and from the per-sample series. The threshold is a compromise: above it descents are usually free-running (on -3% the terminal speed of an 80 kg rider with $C_dA = 0.3\,\text{m}^2$ is about 12 m/s, so no braking is needed) and they carry valuable information on drag with no pedalling noise; below it braking becomes the norm. The effect is quantified in Section 5.3.

4.5 Identifiability

The model has three physically distinct speed dependences: P_{roll} and P_{grav} grow like v, P_{aero} like $v\,v_a^2$, and P_{accel} like $v\dot{v}$. On a ride at constant speed on a flat road these are collinear: a change δC_{rr} is indistinguishable from a proportional change of C_dA. The first-order bias of the estimate caused by an error δP_{other} in any non-aerodynamic term follows from Eq. (10):

$$\delta\widehat{C_dA} \;\approx\; -\frac{\sum_i k_i\,\delta P_{\text{other},i}}{\sum_i k_i^2}, \qquad \delta P_{\text{roll}} = mg\,\delta C_{rr}\,v, \quad \delta P_{\text{grav}} = mg\,\delta s\,v. \tag{12}$$

Equation (12) shows that a grade error δs and a rolling-resistance error δC_{rr} are *equivalent*: an altimeter bias of 0.1 % grade has the same effect as $\delta C_{rr} = 0.001$, and (Table 5) both shift $\widehat{C_dA}$ by $-0.0084\,\text{m}^2$. Only variation of speed over the ride (accelerations, hills of different gradient) separates the terms — the principle behind regression and virtual-elevation methods (Martin et al., 1998; Chung, 2003; Debraux et al., 2011). The present estimator *does not fit* C_{rr}; it fixes it, so any error in C_{rr} is aliased into C_dA.

4.6 Parameters

Table 3 gives all constants of the power-balance model with their provenance.

Constant	Value	Code	Basis
Gravity g	$9.806\,65\,\text{m/s}^2$	`G`	standard value
Bike mass	8 kg	`BIKE_MASS_KG`	typical road bike
C_{rr}	0.005	`CRR`	typical road tyre; *fixed*, not fitted
Drivetrain loss λ	0.03	`DRIVETRAIN_LOSS`	(Martin et al., 1998; Spicer et al., 2001); zero for hub meters
Min. speed for fit	5 m/s	`MIN_CDA_SPEED_MS`	drag signal vs. noise
Max. descent grade	5 %	`MAX_DESCENT_GRADE_PCT`	unmeasured braking, Section 4.4
Speed smoothing	5 samples	`SPEED_SMOOTH_S`	Section 4
Wind height factor k_w	0.7	`WIND_HEIGHT_FACTOR`	boundary layer, open terrain
Valid fit window	$[0.10, 0.80]$	`CDA_VALID`	plausibility
Min. samples	120	(inline)	stability of fit
Per-sample display range	$[0, 1]$	(inline)	plotting only

Table 3. Constants of the power-balance model.

5 Numerical verification

5.1 Recovery of a known C_dA (Monte Carlo)

We constructed synthetic rides from the forward model, Eq. (2), with a known $C_dA = 0.300\,\mathrm{m}^2$, applied measurement noise, and compared the estimators over 300 repetitions.

Ride (a): fast, rolling.
3600 s; speed $v(t) = 9.5 + 2.5\sin(2\pi t/240) + 1.2\sin(2\pi t/71 + 1)$ m/s; grade $2.5\sin(2\pi t/700) + 1.0\sin(2\pi t/190)$ %; mean power 267 W; 72 kg rider + 8 kg bike; 200 m altitude, 18 °C, 50 % RH; heading 90°, wind 4 m/s (at 10 m) from 30° (head-wind component 1.4 m/s at rider height). Noise: power 8 W (white), speed 0.15 m/s (white), grade error 0.25 % (AR(1), 30 s correlation time).

Ride (b): slow, noisy, hilly.
$v(t) = 6.5 + 1.8\sin(2\pi t/200) + 0.8\sin(2\pi t/53)$ m/s (mean 6.5 m/s, part of the ride below the 5 m/s cut-off), grade $3\sin(2\pi t/500)$ %, mean power 85 W, 300 m, 15 °C, 50 % RH, no wind; noise: power 25 W, speed 0.2 m/s, no grade error.

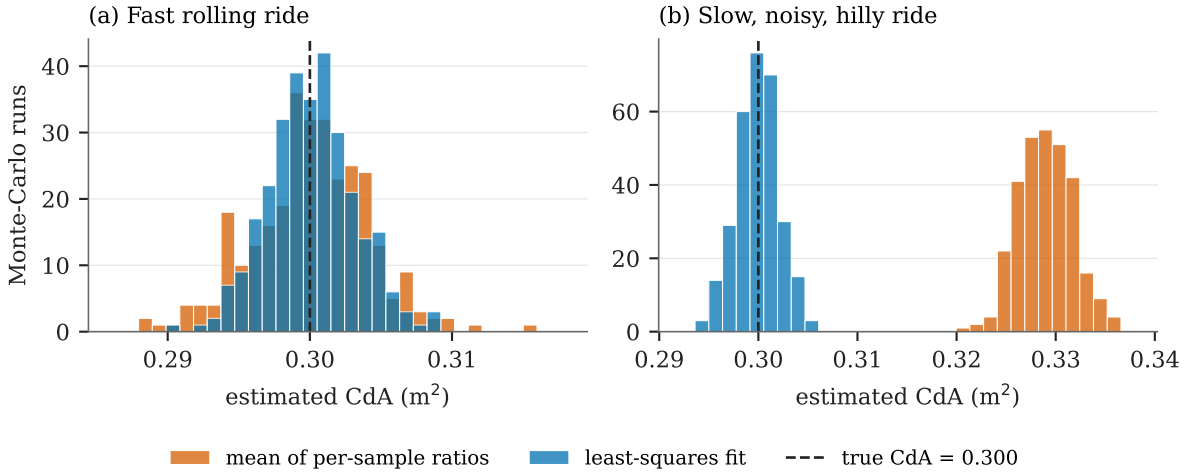


Figure 1. Distribution of the estimated C_dA over 300 Monte-Carlo repetitions for two synthetic rides with true $C_dA = 0.300\,\mathrm{m}^2$. (a) Fast ride: both estimators are unbiased, the least-squares fit has 25% lower spread. (b) Slow, noisy ride: the mean of per-sample ratios is biased by $+0.029$ m^2 ($+10$%) because implausible negative ratios are discarded; the least-squares estimate remains unbiased.

	Ride (a)		Ride (b)	
Estimator	mean	s.d.	mean	s.d.
Least squares, Eq. (10)	0.3001	0.0030	0.2999	0.0021
Mean of per-sample ratios	0.3001	0.0040	0.3290	0.0027

Table 4. Estimated C_dA (m^2); true value 0.300. In the noise-free case the least-squares estimate is 0.3003.

For ride (a), Eq. (11) predicts that white power noise of 8 W alone would give a standard deviation of only $8/\sqrt{\sum k_i^2} = 1.6 \times 10^{-4}$ m^2 ($\sum k^2 = 2.5 \times 10^9$). The observed spread of 0.003 m^2 — twenty times larger — therefore comes almost entirely from the *correlated* errors (grade and speed), which do not average out. This is the central quantitative result for C_dA: *random power noise is irrelevant; systematic errors dominate*.

5.2 Systematic errors

Using the same noise-free ride (a), we perturbed one input at a time and recorded the shift of the fitted value (Table 5, Figure 2).

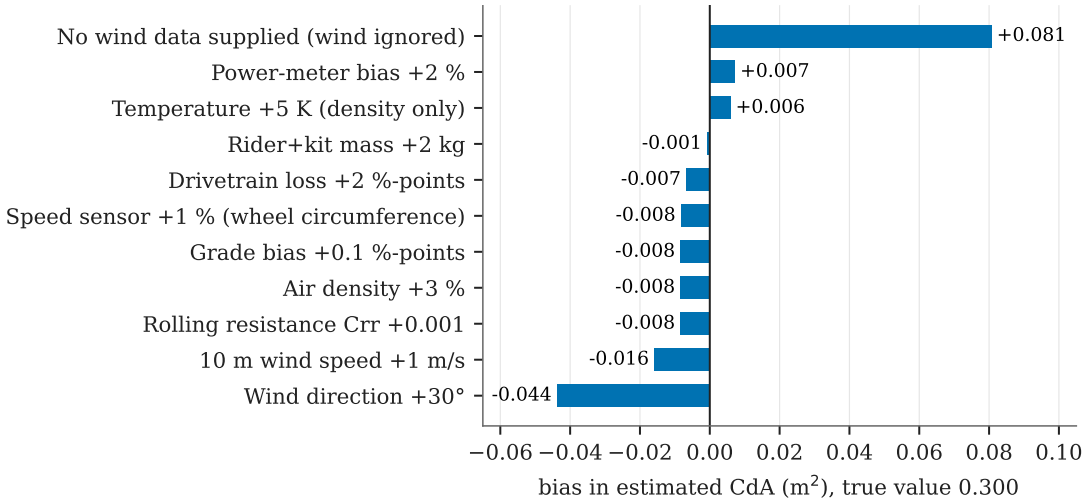


Figure 2. Bias in the fitted C_dA caused by single systematic input errors (noise-free ride (a), true value 0.300 m^2).

Perturbation	$\delta\widehat{C_dA}$ (m^2)	relative
No wind data supplied (wind ignored)	+0.081	+27%
Power-meter gain +2%	+0.007	+2.4%
Temperature +5 K (density only)	+0.006	+2.0%
Mass +2 kg	−0.001	−0.3%
Drivetrain loss +2 percentage points	−0.007	−2.3%
Speed sensor +1%	−0.008	−2.8%
Grade bias +0.1 percentage points	−0.008	−2.8%
Air density +3%	−0.008	−2.8%
C_{rr} + 0.001	−0.008	−2.8%
10 m wind speed +1 m/s	−0.016	−5.3%
Wind direction +30°	−0.044	−14.6%

Table 5. First-order sensitivity of the fitted C_dA (true 0.300 m^2) to single input errors.

Four lessons follow.

1. **Wind is the largest error source.** Omitting the wind altogether in a ride with 1.4 m/s average head wind at rider height inflates C_dA by 27 %; a 30° error in wind direction, easily possible with a modelled hourly value, costs 15 %. The weather data are modelled at a fixed point and hour, not measured on the route; for point-to-point rides along a single heading these errors do not cancel, whereas for loops the first-order effect partially cancels.
2. **Grade, rolling resistance and density produce the same signature** and are mutually degenerate at first order (Eq. (12)). An uncertainty of ±0.001 in C_{rr} and ±0.1 % in mean grade each gives ±0.008 m^2 (±2.8 %).
3. **The sensor calibration matters at the 2 % level**: power-meter gain, wheel circumference (speed) and drivetrain efficiency.
4. **Constant-speed rides are worst.** For a steady 10 m/s ride on flat ground a +0.001 error in C_{rr} moves C_dA by −0.013 m^2, against −0.008 m^2 on the rolling ride, because there is no speed variation to separate rolling from drag.

A plausible total uncertainty, combining errors in C_{rr}, grade, wind (1 m/s, 15°) and calibration in quadrature, is of the order of ± 0.03 m^2 ($\approx 10\%$) for a single ride, and more with poor wind data. This is not negligible against the gain from an aerodynamic position change, so *trends over many rides and comparisons under identical conditions are meaningful, absolute single-ride values should be read with caution*.

5.3 Braking on steep descents

We simulated a ride (physical integration, 3600 s, true $C_dA = 0.300$ m^2, power 250 ± 60 W, grades $\pm 7\%$, no wind, no noise) in which the rider stops pedalling and brakes with a constant 40 N on descents steeper than 5% (865 s, 24 % of the ride, mean speed 7.6 m/s on those samples, mean unmeasured dissipation ≈ 300 W of residual per sample). Without the exclusion the fitted C_dA is 0.346 m^2 ($+0.046$, $+15\%$); with it, 0.304 m^2 ($+0.004$). The remaining $+0.004$ m^2 is leakage of the braking transients through the 5 s smoothing into neighbouring, unfiltered samples. Even though the steep samples carry only 7% of the total fit weight here, they dominate the error, because every one of them is biased in the same direction. A unit test (`test_braking_on_descents_is_not_booked_as_drag`) enforces this behaviour: slowing the rider on steep descents by 20 % must not change the estimate.

6 Limitations of the C_dA model (not corrected in the code)

- **Unmeasured dissipation on moderate descents and in corners.** Steep descents ($< -5\%$) are excluded (Section 4.4), but braking before bends, on moderate gradients or in traffic remains in the data, as do cornering and rubbing losses; each makes $\hat{P}_{\text{aero}}$ too large and *biases* C_dA *upward*. A speed- or curvature-based filter (excluding samples with strong deceleration at $P = 0$) would remove more of it.
- **Drafting and group riding** reduce the effective C_dA by tens of percent (Crouch et al., 2017); a ride in a group does not measure the free-air value.
- **Yaw dependence, posture changes, standing and sprinting** are not modelled (Section 2).
- **Elevation noise.** Grade is taken from the smoothed altitude profile; short, steep errors in barometric or GPS altitude translate directly into P_{grav} errors with the same signature as C_{rr} (Eq. (12)).
- **Tail wind stronger than ground speed** is mis-signed, and the weather input is a coarse hourly model value (Section 2).
- **Constant C_{rr} and λ** are assumed; in reality C_{rr} varies with surface, tyre pressure and (weakly) speed, and λ with power and cadence.

Part II
Two-node model of core and skin temperature

7 Model structure

Human thermoregulation has been described by models of widely differing complexity, from the two-node model of Gagge et al. (1971) to the multi-segment, multi-layer models of Stolwijk (1971) and Fiala et al. (1999) (see Havenith and Fiala, 2016 for a review). BikeToolsAnalyzer uses a *two-node* model: a *core* node (organs, muscle, deep tissue; 90% of the body mass) and a *skin* node (10%), each with a uniform temperature, T_c and T_s (Figure 3). The nodes exchange heat by conduction and blood flow; the skin exchanges heat with the environment by convection/radiation through the clothing and by evaporation of sweat; the core loses heat by

breathing.

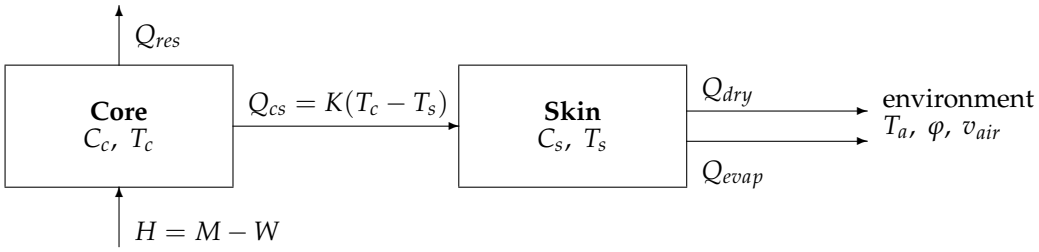


Figure 3. Structure of the two-node model. Heat H is produced in the core; it flows to the skin (Q_{cs}), is lost to the air by dry exchange and evaporation, and by respiration from the core.

The governing equations are the energy balances of the two nodes,

$$C_c \frac{\mathrm{d}T_c}{\mathrm{d}t} = H - Q_{res} - Q_{cs}, \tag{13}$$

$$C_s \frac{\mathrm{d}T_s}{\mathrm{d}t} = Q_{cs} - Q_{dry} - Q_{evap}, \tag{14}$$

with heat capacities $C_c = (1 - f_s)\, m\, c_c$ and $C_s = f_s\, m\, c_s$, skin mass fraction $f_s = 0.1$, $c_c = 3470\,\mathrm{J/(kg\,K)}$, $c_s = 3500\,\mathrm{J/(kg\,K)}$ (the value commonly used in the Gagge model is $\approx 3490\,\mathrm{J/(kg\,K)}$). Adding Eqs. (13)–(14) shows that the transfer term Q_{cs} cancels, so the total stored heat changes by exactly

$$\frac{\mathrm{d}(C_c T_c + C_s T_s)}{\mathrm{d}t} = H - Q_{res} - Q_{dry} - Q_{evap}, \tag{15}$$

a property that the discretisation preserves exactly and that we use as a verification test (Section 11).

7.1 Body surface area

Heat-exchange coefficients are defined per unit surface, so the body surface area A is estimated from height H_{cm} (cm) and mass W (kg) with the Du Bois and Du Bois (1916) formula,

$$A = 0.007184\; H_{cm}^{0.725}\, W^{0.425} \quad (\mathrm{m^2}), \tag{16}$$

which gives $1.60\,\mathrm{m^2}$ for $165\,\mathrm{cm}/55\,\mathrm{kg}$, $1.85\,\mathrm{m^2}$ for $175\,\mathrm{cm}/70\,\mathrm{kg}$ and $2.09\,\mathrm{m^2}$ for $185\,\mathrm{cm}/85\,\mathrm{kg}$. The formula is derived for upright standing adults; the effective exchange area on the bicycle (crouched posture, parts of the body shielded) is smaller, which is not corrected for.

8 Heat production

Let $W \geq 0$ be the mechanical (crank) power. The metabolic rate is

$$M = \max\!\left(\frac{W}{\eta},\; M_0 A\right), \qquad H = M - W, \tag{17}$$

with $M_0 = 58.2\,\mathrm{W/m^2}$ the resting metabolic rate (1 met, International Organization for Standardization, 2005) and $\eta = 0.23$ the *gross* efficiency. Only the fraction η of the metabolic energy leaves the body as mechanical work; the remainder, $H = W(1/\eta - 1) \approx 3.35\,W$, is heat. Reported gross efficiencies of trained cyclists lie in roughly the range 18–25% depending on power, cadence

and fitness (Ettema and Lorås, 2009). The max ensures the rider produces at least the resting heat when freewheeling (without it, an idle rider cooled indefinitely; this was one of the defects corrected during the audit).

Two simplifications deserve mention: (i) η is a constant, whereas it varies with power and with the oxygen-uptake slow component (Ettema and Lorås, 2009); (ii) M responds instantaneously to power, whereas oxygen uptake lags power by about 30–60 s. Both are minor next to the sensitivity to η itself (Section 11.5).

9 Heat exchange with the environment

9.1 Dry exchange through the clothing

Sensible heat (convection and radiation combined) flows through the series resistance of the clothing and the air boundary layer:

$$Q_{dry} = \frac{A\,(T_s - T_a)}{R_{cl} + R_{air}(v_{air})}, \qquad R_{air} = \frac{1}{h_a}, \quad h_a = \max\!\left(\alpha\, v_{air}^{\,n},\; h_0\right), \tag{18}$$

with $\alpha = 8.5$, $n = 0.84$, $h_0 = 7.8\,\mathrm{W/(m^2K)}$ and v_{air} the *apparent air speed* $\sqrt{v_a^2 + v_\perp^2}$ of Eq. (3) computed with the smoothed ground speed and the wind at rider height (identical to the wind treatment of the drag model; the thermal model previously computed the wind separately). Clothing insulation is

$$R_{cl} = 0.155 \sum_j I_j \quad (\mathrm{m^2\,K/W}), \tag{19}$$

where I_j (in clo; $1\,\mathrm{clo} = 0.155\,\mathrm{m^2K/W}$, Gagge et al., 1941) is the nominal insulation of each garment from the catalogue (e.g. summer jersey 0.15 clo, long-sleeve jersey 0.25 clo, rain jacket 0.35 clo). Summing garment values is the standard first approximation of ISO 9920 (International Organization for Standardization, 2007).

Assessment. The coefficient $h_a = 8.5\,v^{0.84}$ is an empirical combined (convective+radiative) coefficient, not a standard correlation. Figure 4 compares it with the ASHRAE correlation for a clothed standing person, $h_c = 8.3\,v^{0.6}$ ($0.2 < v < 4\,\mathrm{m/s}$, ASHRAE, 2017), to which a linearised radiative coefficient $h_r \approx 4.7\,\mathrm{W/(m^2K)}$ must be added. The model is lower than the standard value at $v \lesssim 3\,\mathrm{m/s}$ and increasingly higher above it ($33\,\mathrm{W/(m^2K)}$ against 26.5 at $5\,\mathrm{m/s}$; 59 against 38 at $10\,\mathrm{m/s}$, Table 6). At cycling speeds the standard correlation is outside its validated range ($\leq 4\,\mathrm{m/s}$), so a stronger dependence is not unreasonable: however, the exponent and prefactor have not been calibrated against measurements for cyclists. The consequence is shown in Section 11.5: the model is moderately insensitive to α at moderate temperatures ($\pm 20\%$ in α changes the final core temperature by $\lesssim 0.15\,\mathrm{K}$).

Still-air limit. A pure power law has a serious shortcoming at low speed: $h_a \to 0$ as $v_{air} \to 0$, whereas radiation and natural convection guarantee a floor of about $4.7 + 3.1 \approx 7.8\,\mathrm{W/(m^2K)}$ even in still air (ASHRAE, 2017). The unfloored model gave $h_a = 2.2\,\mathrm{W/(m^2K)}$ at $0.2\,\mathrm{m/s}$, which kept riders too warm at stops, on slow climbs and at rest (skin $\approx 2\,\mathrm{K}$ too warm; finding O1). The coefficient is therefore $h_a = \max(\alpha v_{air}^{\,n},\, h_0)$ with $h_0 = 7.8\,\mathrm{W/(m^2K)}$; the floor is active only below $\approx 0.9\,\mathrm{m/s}$ and leaves all riding conditions unchanged.

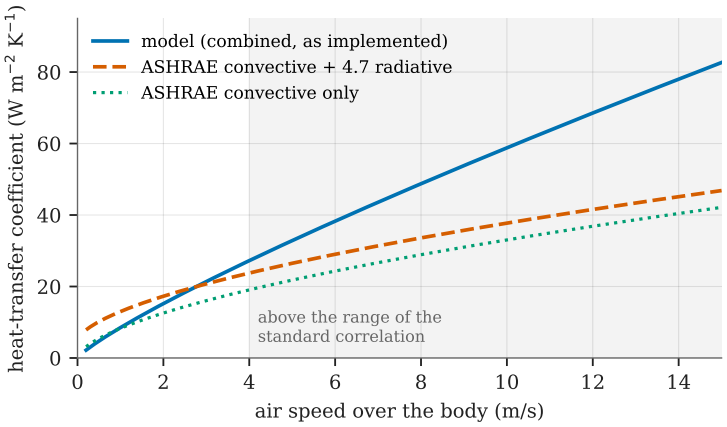


Figure 4. Heat-transfer coefficient used by the model compared with the ASHRAE correlation for a clothed person (ASHRAE, 2017).

v_{air} (m/s)	model $8.5\,v^{0.84}$	ASHRAE $8.3\,v^{0.6}$	ASHRAE $+\,4.7$
1	8.5	8.3	13.0
2	15.2	12.6	17.3
5	32.9	21.8	26.5
10	58.8	33.0	37.7

Table 6. Coefficients in W/(m^{2}K).

Clothing under wind. Insulation measured on a still manikin decreases substantially with wind and movement through air penetration and the pumping effect, by up to about 40–50 % in thin ensembles (Havenith et al., 1990). The model applies the nominal static value; for a rider moving at 10 m/s this overestimates the insulation of non-windproof garments and underestimates it for windproof ones (the catalogue gives a rain jacket 0.35 clo regardless of air speed). This is open finding O6.

9.2 Evaporative heat loss

Evaporation is the dominant heat sink when the body must dissipate heat above what the skin can lose dry. Two quantities limit it: the *sweat supply* S that the body secretes, and the *evaporative capacity* E_{max} that the environment allows. In the physical treatment of Gagge and Nishi (1977), $E_{max} = h_e\,(p_{s,sk} - p_a)\,A$ with the skin-surface saturation pressure $p_{s,sk}$ and the ambient vapour pressure p_a, and the evaporative coefficient h_e related to the convective one by the Lewis relation ($\approx 16.5\,\mathrm{K/kPa}$). Our model uses a phenomenological version of this structure:

$$S = g_s\,\max(0,\,T_c - 37), \qquad g_s = 1000\,\mathrm{W/K}, \tag{20}$$

$$E_{max} = \varepsilon_0\,\max(v_{air}, 0.2)^{0.3}\,A\,\max\!\left(0,\,\frac{p_s(T_s) - \varphi\,p_s(T_a)}{p_s(33\,^\circ\mathrm{C})}\right), \qquad \varepsilon_0 = 150\,\mathrm{W/m^2}, \tag{21}$$

$$Q_{evap} = \min(S,\,E_{max}), \tag{22}$$

with $p_s(T) = 0.6112\,\mathrm{kPa}\,\exp[17.67\,T/(T + 243.5)]$ (Bolton, 1980). The bracket in Eq. (21) is the vapour-pressure gap between wet skin and ambient air, normalised to a skin at 33 °C in dry air; it reduces to $1 - \varphi$ when air and skin are at the same temperature, which is the form the model used before the audit. Using $1 - \varphi$ alone, however, made the evaporative capacity independent of temperature, which is wrong: cool humid air (10 °C, 90 %) has a vapour-pressure gap of 0.78 of the dry value for typical skin temperatures, whereas $1 - \varphi$ gives 0.10. Eq. (21) corrects

this. The $\max(v_{air}, 0.2)$ floor prevents the capacity from vanishing in still air (physical body movement and natural convection always provide some air motion).

Assessment. The sweat supply, Eq. (20), has the right order of magnitude — 1000 W corresponds to about 1.5 L/h of fully evaporated sweat (latent heat $\approx 2430\,\mathrm{kJ/kg}$), within the range reported for athletes — but it is driven only by core temperature, and not, as in Gagge-type controllers, by a weighted combination of core and skin temperatures. E_{max} is *not* derived from the Lewis relation: at 1 m/s and dry air, $\varepsilon_0 = 150\,\mathrm{W/m^2}$ is several times lower than the nude-skin value ($\approx 16.5 \times 8.3 \times 5.03 \approx 690\,\mathrm{W/m^2}$), which is plausible as an effective value for a clothed rider (clothing vapour resistance, sweat dripping rather than evaporating) but is a calibration constant. The speed exponent of 0.3 is likewise smaller than the exponent of the convective coefficient. Neither has been fitted to data (open finding O4).

9.3 Respiratory heat loss

The sensible and latent heat lost through breathing is, in the standard formulation of Fanger (1970) and International Organization for Standardization (2005),

$$Q_{res} = \max\bigl(0,\ 0.0014\,M\,(34 - T_a) + 0.0173\,M\,(5.87 - p_a)\bigr), \tag{23}$$

with M in W and the ambient vapour pressure $p_a = \varphi\, p_s(T_a)$ in kPa. The two terms are the dry and the latent loss, both proportional to ventilation, which scales with M. At 200 W ($M = 870$ W) in 20 °C, 50 % RH the formula gives 88 W, about 13 % of the heat produced.

9.4 Core–skin transfer and vasodilation

The conductance between core and skin combines tissue conduction with convective transport by skin blood flow:

$$K = A\left[k_{min} + k_v\,\frac{\sigma}{1+\sigma}\right], \qquad \sigma = \max(0,\ T_c - 36.8), \tag{24}$$

$$k_{min} = 12.6\,\mathrm{W/(m^2K)} = 5.28\ (\text{tissue}) + 7.3\ (\text{skin blood flow}), \tag{25}$$

$$k_v = 150\,\mathrm{W/(m^2K)}. \tag{26}$$

The basal terms 5.28 and 7.3 are those of the Gagge model (Gagge et al., 1971; Gagge and Nishi, 1977) (minimal skin blood flow of $6.3\,\mathrm{L/(m^2h)}$, multiplied by the volumetric heat capacity of blood, $1.163\,\mathrm{W\,h/(L\,K)}$). The saturating vasodilation term is a heuristic: it rises smoothly with the core temperature above 36.8 °C, avoids a hard switch, and reaches half its asymptote at $T_c = 37.8$ °C. In Gagge-type models skin blood flow increases linearly with the core-temperature error up to a physiological cap of about $90\,\mathrm{L/(m^2h)}$, i.e. $\approx 105\,\mathrm{W/(m^2K)}$; the present form reaches that conductance only at $\sigma \approx 1.6$ ($T_c \approx 38.4$ °C) — later and more gradually than the physiological controller — so it is, if anything, *conservative* regarding heat dissipation. It also ignores the inhibition of skin vasodilation by cold skin and the reduction of blood flow to the skin during heavy exercise.

9.5 Exercise set point

In exercising humans the steady-state core temperature is not fixed by the environment but rises in proportion to the *relative* workload (fraction of maximal oxygen uptake), almost independently of ambient temperature over a wide range (Saltin and Hermansen, 1966): the thermoregulatory set point is shifted upward during exercise, so that thresholds for sweating and skin vasodilation

are higher than at rest. A model with fixed resting thresholds (37.0 °C and 36.8 °C) holds the core at about 37.1 °C at 200 W in mild air, ~ 0.8 K below the measured plateau. We shift both thresholds by

$$\Delta(t) = g_{sp}\,\phi(t), \qquad \frac{\mathrm{d}\phi}{\mathrm{d}t} = \frac{\min\big(W(t)/\mathrm{FTP},\ \phi_{max}\big) - \phi}{\tau_{sp}}, \tag{27}$$

so that sweat onset is $37.0 + \Delta$ and vasodilation onset $36.8 + \Delta$ (°C), with $g_{sp} = 1.0$ K per unit of FTP, $\phi_{max} = 1.2$ and $\tau_{sp} = 300$ s (the first-order filter reproduces the observed rise of core temperature over the first 10–20 min). The relative intensity is taken as power over the rider's FTP, because the FTP is known to the application while $\dot{V}O_{2,max}$ is not; FTP is a proxy for the lactate threshold, at about 75–85% of $\dot{V}O_{2,max}$ in trained riders. Without an FTP the shift is zero and the model reduces to the resting-threshold version. The gain g_{sp} is chosen to reproduce a core temperature of ≈ 38 °C at about 80% of FTP in mild air; it has *not* been fitted to data, and its effect is shown in Section 11.4.

10 Numerical integration

Equations (13)–(14) are integrated with the explicit (forward) Euler method with time step Δt. For a linear relaxation with time constant τ, forward Euler is stable only for $\Delta t < 2\tau$, and monotone for $\Delta t < \tau$ (Hairer et al., 1993). The skin node is the stiffest: $\tau_s = C_s/(K + A/R_{tot})$ ranges from 1050 s (resting conductance, no air flow) to 71 s (full vasodilation, 10 m/s air speed), computed for a 70 kg, 175 cm rider ($C_s = 24.5$ kJ/K). The stability limit is therefore $\Delta t < 141$ s. For 1 Hz data this holds with a large margin, but *recording gaps and smart-recording files with long intervals could violate it*: we reproduced, with $\Delta t = 200$ s, non-physical skin temperatures of 16 °C and jumps of 36 K between samples (the original code skipped only intervals above 300 s). The integration now subdivides every interval into sub-steps of at most 5 s, a factor of ≈ 14 below the minimum τ_s. Table 7 shows convergence: results are identical for sampling intervals of 1 and 5 s, and differ by less than 0.1 K even for 299 s, the residual being due to the sample-and-hold of the inputs rather than to the integrator.

Sampling interval (s)	core T_c at 60 min (°C)	skin T_s at 60 min (°C)
1	39.526	36.47
5	39.526	36.47
60	39.508	36.45
200	39.461	36.41
299	39.423	36.37

Table 7. Step-size convergence for 250 W, 30 °C, 50 % RH, 8 m/s, summer jersey, after the sub-stepping fix. (Before the fix, the 200 s and 299 s runs diverged.)

Initial state. Starting the integration at an arbitrary skin temperature produces a fictitious warm-up transient (the original code used $T_c = 36.0$, $T_s = 32.0$ °C). The ride now starts with a warm, rested rider, $T_c = 36.9$ °C (`CORE_START_C`), and a skin temperature in quasi-steady balance with the first sample's air and clothing in still air,

$$T_{s,0} = \frac{k_{min} A\, T_{c,0} + g\, T_a}{k_{min} A + g}, \qquad g = \frac{A}{R_{cl} + 1/h_0}. \tag{28}$$

An earlier version integrated a "resting equilibrium" with the same equations; with a realistic still-air coefficient that equilibrium is cold at low ambient temperatures, because the model has no shivering thermogenesis (Section 12), and so is not a meaningful starting condition.

11 Verification and behaviour

11.1 Energy conservation

Summing Eq. (15) over the simulation shows that the heat stored in the two nodes must equal the integral of generated minus lost heat. This is a pure check on the code, independent of the physics; it holds to the numerical precision of the scheme and is enforced by a unit test (`test_energy_is_conserved`).

11.2 Resting state

Table 8 shows a rider who sits still after the starting state of Section 10 (no power, no air movement). A resting rider produces $M_0 A = 108\,\mathrm{W}$. At 30 °C the core is stable (36.9 → 37.0 °C) and skin settles near 35.7 °C, as in a warm environment. In cooler air the rider *loses more heat than a resting body produces*: with only a jersey the core falls by 0.6 K in an hour at 20 °C and by 1.9 K at 5 °C and 1.4 K at 10 °C with two layers. Real riders would shiver (raising M by a factor of 2–3) and vasoconstrict; the model has neither, so **long stops in cold air are overcooled** (finding O9). The change from the previous version, in which the resting skin was about 2 K too warm and the core was pinned between the controller thresholds, is the consequence of the still-air floor of Section 9.1.

T_a (°C)	clothing	core, start	core, after 60 min	skin, after 60 min
5	jersey + long-sleeve jersey	36.90	35.01	26.4
10	jersey + long-sleeve jersey	36.90	35.54	28.2
20	summer jersey	36.90	36.31	30.8
30	summer jersey	36.90	37.03	35.7

Table 8. Resting rider (power 0, no air movement), temperatures in °C.

11.3 Exercise scenarios

Table 9 and Figure 5 show one-hour rides at constant power on flat ground (8 m/s, 50 % RH, rider 70 kg/175 cm, FTP 250 W, no wind).

Power	Conditions	core, 60 min	skin, 60 min	mean H (W)	mean heat lost (W)
150 W	20 °C, jersey	37.55	30.3	502	468
200 W	10 °C, 3 layers	37.97	32.3	670	586
200 W	20 °C, jersey	37.90	31.3	670	607
200 W	30 °C, jersey	38.68	35.2	670	557
300 W	25 °C, jersey	39.55	35.1	1004	828

Table 9. Temperatures in °C. The "3 layers" are summer jersey, long-sleeve jersey and gilet. Mean heat lost is below or close to production because the core is storing heat while the set point rises; the skin cools from its starting value to the exercise value during the first minutes.

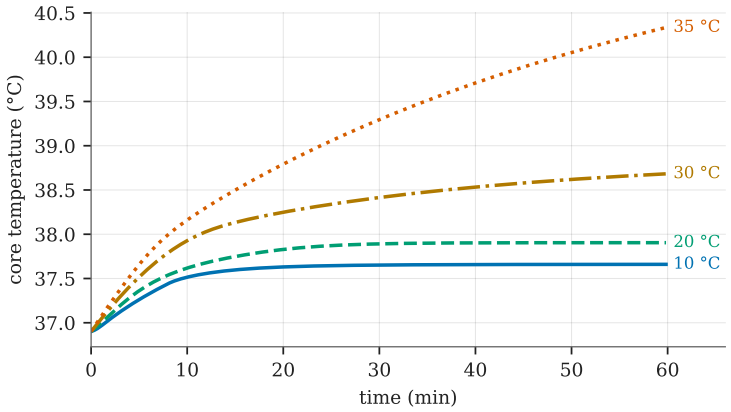


Figure 5. Core temperature during a 60-min ride at 200 W (summer jersey, 8 m/s, 50 % RH). In cool and mild air the core temperature rises over the first 10–20 min to a plateau of about 37.7–37.9 °C, as the exercise set point builds up; at 30 °C it keeps rising, and at 35 °C it reaches 40 °C. FTP 250 W.

The qualitative behaviour is that of a thermoregulating body: in cool conditions heat is lost as fast as it is produced (the core stays near 37 °C); as the temperature rises, the dry gradient disappears and the core must store heat until sweating (which starts above 37 °C plus the exercise set-point shift, Section 9.5) and vasodilation balance production. At 35 °C and 60 % RH the model predicts $T_c \approx 40.8$ °C after an hour. Experimental studies place exhaustion in hyperthermic exercise at a core temperature of about 40 °C (González-Alonso et al., 1999); the model has no behavioural feedback (the rider cannot slow down), no dehydration and no cardiovascular limit, so such values should be read as an upper bound on heat storage rather than a forecast.

The effect of humidity is quantified in Figure 6. At 220 W and 28 °C, increasing relative humidity from 20 to 95 % raises the one-hour core temperature from 38.3 to 39.6 °C; at 24 °C the same change causes only 0.3 K, because evaporation is not yet the limiting process. The dependence on RH is monotone in every case, as it must be physically (this is a regression test).

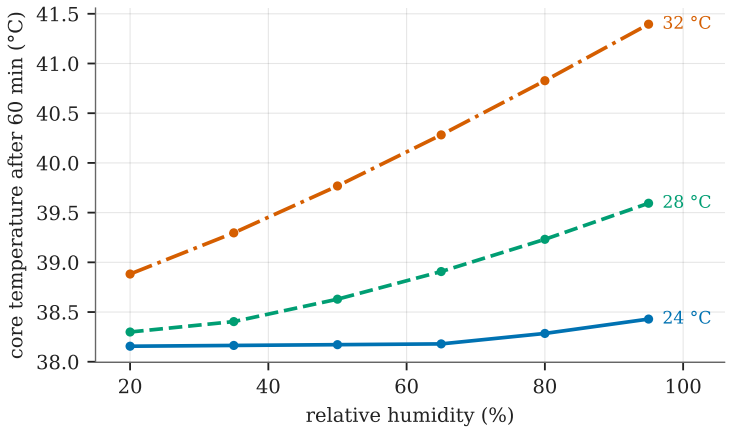


Figure 6. Core temperature after 60 min at 220 W as a function of relative humidity, for three ambient temperatures.

11.4 Effect of the exercise set point

Table 10 compares one hour at 200 W (same conditions as above) without the set-point shift, i.e. with the fixed resting thresholds, and with the shift for riders of different FTP. Without it the model predicts about 37.1 °C in 20 °C air, whereas the measured plateau at a moderately

high relative workload is ≈ 38–39 °C (Saltin and Hermansen, 1966); with FTP 250 W (200 W = 80% of FTP) the prediction is 37.9 °C. A stronger rider (FTP 400 W, 50%) is less stressed by the same power (37.6 °C), a weaker one (FTP 200 W, 100%) more (38.1 °C), as in the experimental literature. The effect decreases in hot air (+0.5 K at 30 °C against +0.8 K at 20 °C for FTP 250 W), because the heat-storage regime is then limited by the environment, not by the set point. The gain g_{sp} is an uncalibrated choice: ±20% of it changes the final core temperature by ≈ ±0.13 K (Figure 7).

	10 °C (3 layers)	20 °C	30 °C
Fixed resting thresholds (no FTP)	36.87	37.13	38.22
FTP 400 W (50 %)	37.36	37.61	38.49
FTP 250 W (80 %)	37.66	37.90	38.68
FTP 200 W (100 %)	37.86	38.10	38.82

Table 10. Core temperature (°C) after 60 min at 200 W by rider FTP. The 10 °C column uses jersey, long-sleeve jersey and gilet; the others a summer jersey.

11.5 Parameter sensitivity

Each constant was varied by ±20% (clothing insulation values of all garments scaled together) in two scenarios: a warm one (220 W, 28 °C, 50 %) and a cool one (180 W, 12 °C, 70 %, two layers), both with FTP 250 W. Figure 7 shows the change in core temperature at 60 minutes.

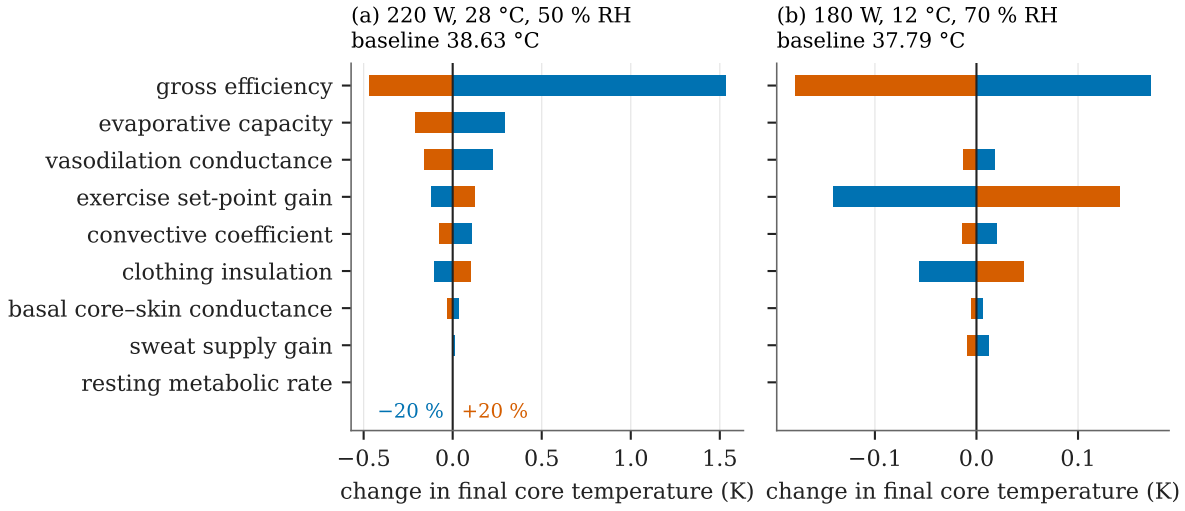


Figure 7. Sensitivity of the final core temperature to ±20% changes of each model parameter (blue: −20%; orange: +20%). Baselines are 38.63 °C (warm) and 37.79 °C (cool); FTP 250 W.

- The result in warm conditions is dominated by the **gross efficiency** η: $\eta = 0.184$ (−20%) raises the final core temperature by 1.5 K and $\eta = 0.276$ lowers it by 0.5 K. The asymmetry arises because the heat produced is $P(1/\eta - 1)$, which changes by +32% and −22% respectively. Since individual gross efficiency varies at least over 0.18–0.25 (Ettema and Lorås, 2009), *the largest uncertainty of the predicted core temperature is the rider's own efficiency*.
- The evaporative capacity (+0.29 / − 0.21 K) and the vasodilation conductance (+0.23 / − 0.16 K) are next. The set-point gain (±0.13 K), the clothing insulation (±0.10 K) and the convective coefficient ($\lesssim$ 0.11 K) matter less.
- In the cool scenario no sweat is secreted and all sensitivities are $\lesssim$ 0.2 K (η: +0.17 / − 0.18 K; set-point gain ±0.14 K).

- The resting metabolic rate has no effect during exercise, as expected from Eq. (17) ($W/\eta \gg M_0 A$ once $W > 25\,\mathrm{W}$).

12 Limitations of the thermal model

- **No calibration or validation against measured core temperature** (e.g. with an ingestible pill or a skin-core sensor). The structure and most basal constants come from the literature; the coefficients α, n, ε_0, the exponent 0.3, g_s, k_v, the thresholds and the set-point gain g_{sp} are heuristic and unvalidated (the gain is chosen to give $\approx 38\,^\circ\mathrm{C}$ at 80% of FTP).
- **No cold response.** There is no shivering thermogenesis or cold-induced vasoconstriction: the core of a rider who stops for a long time in cold air falls without limit (Section 11.2), and skin temperatures in cold air during rides are likely too low.
- **Set point from FTP.** The exercise set point is driven by power over FTP, a proxy for relative intensity; it knows nothing of fitness, fatigue or heat acclimatisation, and reacts to the instantaneous power through a single 300 s filter.
- **Solar and radiant load** (direct sun, ground radiation) is not modelled; it is several hundred W in full sun at midday and shifts the dry heat balance substantially in summer.
- **No state for hydration, acclimatisation, circadian phase or sex differences.** Sweat supply is a single linear function of core temperature.
- **Uniform surface:** one skin temperature, one clothing resistance and one air speed for the whole body; helmet, bare head and hands, and the posture-dependent exposed area are ignored.
- **Clothing:** additive static insulation, without air-layer, fit or wind-penetration effects (Section 9.1); no moisture accumulation or wetting of the garments.
- **Output semantics:** `heat_generated_W` and `heat_lost_W` report the values of the *last* sub-step of each sample (relevant only for sampling intervals above 5 s).

Part III
Audit summary

13 Findings register

Table 11 lists the issues found during the audit of both models. "Fixed" means corrected in the code together with a regression test (power balance: pull request #25; thermal model: pull request #26 and the follow-up described in this report).

Table 11. Findings register. A = power balance/$C_d A$; T = thermal, fixed; O = limitations found; the status column says whether they have since been fixed.

ID	Finding	Consequence	Status
A1	Drag-power form $\frac{1}{2}\rho C_d A v_a^2 v$ and the wind-direction convention (direction the wind blows *from*) corrected and tested	systematic $C_d A$ error with wind	Fixed
A2	Mean of per-sample ratios, clipped to a window, used as ride $C_d A$	upward bias up to +10% in noisy rides	Fixed (least squares)
A3	Air density: ISA exponent and humidity (Magnus)	few-percent error in ρ	Fixed
A4	Tail wind stronger than ground speed has the wrong sign (v_a^2 vs. $v_a\lvert v_a\rvert$)	rare; wrong P_{aero} sign in gales	Open

ID	Finding	Consequence	Status
A5	Braking on steep descents booked as drag; fit weights ($\propto v^6$) concentrate on fast descents	upward bias (+15% in the test ride)	Fixed (descents $< -5\%$ excluded)
A8	Braking on moderate grades, in bends and in traffic remains in the data	small upward bias	Open
A6	Constant C_{rr}, λ, bike mass; C_{rr} not fitted	aliased into C_dA (± 0.008 m^2 per 0.001)	Documented
A7	Yaw dependence of C_dA and drafting ignored	effective, not free-air, C_dA	Documented
T1	No resting heat production ($M = W/\eta$ only); idle rider cooled to ≈35.5 °C	wrong during descents/stops	Fixed
T2	Core–skin conductance omitted basal blood flow (5.28 instead of 12.6 W/m^{2}K)	core–skin heat transfer too low	Fixed
T3	Evaporative driving force $1 - \varphi$ instead of the vapour-pressure gap	cool humid air wrongly blocked evaporation	Fixed
T4	Evaporative capacity $\propto v^{0.3} = 0$ in still air	no evaporation at rest/stops	Fixed (floor 0.2 m/s)
T5	Hard-coded initial state 36.0/32.0 °C	fictitious warm-up transient	Fixed (rest state)
T6	Wind computed separately from, and not consistently with, the drag model (now shared through `apparent_air_speed`)	inconsistent air speed between models	Fixed
T7	Forward Euler unstable for sampling intervals $\gtrsim 140$ s	non-physical output for sparse files	Fixed (≤ 5 s sub-steps)
O1	$h_a \propto v^{0.84} \to 0$ at low speed; no still-air floor	skin 2 K warm at rest; slow riding	Fixed ($h_0 = 7.8$)
O2	No exercise-induced set-point shift	core ≈ 0.8 K too low in mild conditions	Fixed (FTP-driven shift, gain uncalibrated)
O3	Fixed gross efficiency $\eta = 0.23$	$\pm 1.8/-0.8$ K at ±20% in heat	Documented
O4	Evaporative capacity not from the Lewis relation; clothing vapour resistance and dripping lumped	uncalibrated	Open
O5	No solar or radiant load	underestimates summer heat storage	Open
O6	Clothing insulation static and additive; no wind penetration	over-/underestimate by up to ∼40%	Open
O7	Convection exponent/prefactor uncalibrated (higher than ASHRAE above 3 m/s)	overestimates cooling at speed	Open
O9	No shivering or cold vasoconstriction	long stops in cold air overcool; core -1.9 K in 1 h at 5 °C	Open
O8	Two Magnus parameter sets (Alduchov–Eskridge in density; Bolton in thermal)	≈ 0.2% difference in e_s; negligible	Documented

14 Recommendations

1. **Calibrate before trusting absolute values.** Collect a small number of rides with a core-temperature sensor and fit η (or a per-rider efficiency), ε_0, k_v and the thresholds; this is the single most valuable step for the thermal model.
2. **Fit the set-point gain** g_{sp} and time constant τ_{sp} of Eq. (27) together with η, and move the exponent/prefactor of the convective coefficient towards the standard $8.3\,v^{0.6} + 4.7$ unless rides show otherwise.
3. **Add a minimal cold response** (shivering thermogenesis and cold vasoconstriction) so that long stops in cold air are not overcooled.
4. **For C_dA: fit C_{rr} and C_dA jointly** on rides with speed variation (regression/virtual-elevation

approach, Martin et al., 1998; Chung, 2003), also exclude samples with strong deceleration at zero power (bends, braking on moderate grades), and report an uncertainty interval from Table 5 alongside the point estimate.

5. **Use measured on-route wind where available** (an anemometer or a nearby station), since the modelled wind dominates the uncertainty of single-ride C_dA.

A Parameters of the thermal model

Constant	Value	Code	Provenance
c_c (core)	3470 J/(kg K)	`C_BODY`	tissue specific heat; ≈ Gagge 3490
c_s (skin)	3500 J/(kg K)	`C_SKIN`	as above
f_s skin mass fraction	0.10	`SKIN_MASS_FRACTION`	Gagge two-node (Gagge et al., 1971)
η gross efficiency	0.23	`EFFICIENCY`	literature range 0.18–0.25 (Ettema and Lorås, 2009)
M_0 resting metabolism	58.2 W/m^2	`M_REST_W_PER_M2`	1 met, ISO 7730
k_{min}	12.6 W/(m^{2}K)	`K_MIN_W_PER_M2K`	Gagge: 5.28 + 7.3
k_v	150 W/(m^{2}K)	`K_VASO_MAX_W_PER_M2K`	*heuristic*
vasodilation onset	36.8 °C	`VASO_ONSET_CORE_C`	*heuristic*
sweat onset	37.0 °C	`SWEAT_ONSET_CORE_C`	*heuristic*
g_s sweat gain	1000 W/K	`SWEAT_GAIN_W_PER_K`	order of magnitude (≈ 1.5 L/h at 1.5 K)
ε_0 evaporative capacity	150 W/m^2	`EVAP_COEFF_W_PER_M2`	*calibration*, not Lewis
evaporation exponent	0.3	`EVAP_EXP`	*heuristic*
reference skin temperature	33 °C	`SWEAT_REF_SKIN_C`	normalisation
minimum air speed	0.2 m/s	`MIN_AIR_SPEED_MS`	natural convection
α, n	8.5, 0.84	`H_CONV_COEFF, H_CONV_EXP`	*empirical*, see Section 9.1
h_0 still-air floor	7.8 W/(m^{2}K)	`H_STILL_AIR_W_PER_M2K`	radiation 4.7 + natural convection 3.1 (ASHRAE, 2017)
Start core temperature	36.9 °C	`CORE_START_C`	warm, rested rider
g_{sp} set-point gain	1.0 K per FTP	`SETPOINT_GAIN_K`	*calibration* to ≈ 38 °C at 80 % FTP (Saltin and Hermansen, 1966)
ϕ_{max}	1.2	`SETPOINT_MAX_FTP_FRACTION`	*heuristic*
τ_{sp}	300 s	`SETPOINT_TAU_S`	*heuristic*
clothing	0.155 m^{2}K/W per clo	(inline)	(Gagge et al., 1941; International Organization for Standardization, 2007)
max. sub-step	5 s	`MAX_SUBSTEP_S`	stability, Section 10

Table 12. Constants of the thermal model.

B Reproducibility

The report, its figures and all numerical results are generated by the scripts `experiments.py` (Monte Carlo, sensitivity and thermal scenarios), `mc_hard.py` (ride (b)) and `figs.py` (figures) archived in `docs/physics_report/` of the repository; random numbers use fixed seeds. The unit tests in `tests/test_aerodyn.py` and `tests/test_thermal.py` encode the properties verified here (exclusion of steep descents, apparent wind at rider height, cross wind, head vs. tail wind, still-air floor, start state, exercise set point, humidity monotonicity, evaporation in cool humid air, energy conservation and stability for sparse sampling).

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